Frontier methods for digital soil mapping, evaluated honestly

A reproducible benchmark on real WoSIS Cerrado profiles

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The problem

- Digital soil mapping now has many "frontier" methods: causal inference, deep learning, graph networks, quantum kernels.
- They are rarely compared under one protocol, and their predictions are rarely reported with calibrated uncertainty.
- So a simple question usually goes unanswered: is any of them actually better than a random forest?

A conceptual worry: causality

- Variable importance in a predictive model is routinely read as evidence that a covariate *causes* the soil property.
- Zhang and Wadoux (2026) argue causal inference in DSM needs conditions that observational soil data rarely meet.
- Tempting shortcut: can a learned representation act as an instrument and rescue causal identification? We test it directly.

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- One R framework, ten method families.
- Every method returns the same object – a posterior.
- One calibration scorer for all of them: CRPS, interval coverage, width.
- That common currency is what makes a like-for-like comparison possible.

The test

- 1095 real WoSIS topsoil organic-carbon profiles, Brazilian Cerrado.
- Covariates from WorldClim, SoilGrids, Copernicus tree cover.
- Spatially blocked cross-validation, matched ensemble budgets.
- Scored against the achievable null: each fold's training-fold mean.

Result 1 – the benchmark

Method	RMSE	R^2
OLS (DAG covariates)	0.583	0.067
Random forest	0.589	0.051
Bayesian spatial	0.597	0.041
Foundation + forest	0.613	-0.003
Graph attention	0.621	-0.080
Quantum kernel	0.663	-0.184

Lower RMSE better; R^2 above 0 beats the null.

Result 1 – what it says

- A plain OLS on DAG-selected covariates is the most accurate and best-calibrated method.
- The three deep or exotic methods fall *below* the achievable null.
- The quantum kernel is the least accurate, and about 300 times slower.
- Elaboration did not pay, here.

Result 2 – causal associations

- Backdoor adjustment returns *associations with sensitivity bounds*, not identified causal effects.
- Precipitation and clay: robust and significant (robustness value 0.16-0.18).
- Tree cover: naive 2.86, adjusted 0.35 once climate is controlled for.
- The raw tree-cover "effect" was mostly shared climatic confounding.

Result 3 – a negative IV result

- We trained a real foundation-model encoder and used its embedding as an instrument for precipitation.
- The instruments are very strong: first-stage F about 4630.
- But they fail the validity test: Sargan p about 3e-13.
- A representation that captures everything cannot satisfy the exclusion restriction. It is the wrong tool.

What it means

- Not "deep learning cannot work" – the sample and signal are small, and a linear model on good covariates is hard to beat at this scale.
- A matched, calibration-aware protocol *shows* this; an accuracy-only, single-method report hides it.
- Backdoor adjustment is useful, but honest about what it returns.
- The learned-representation shortcut to identification is exactly wrong.

Conclusion

- The contribution is not a new best predictor.
- It is a reproducible apparatus for asking *which* method helps – and the discipline to say plainly when none does.
- Every number in this talk is regenerated from the bundled data.

github.com/HugoMachadoRodrigues/edaphos